The National College
Autonomous
Basavanagudi, Bengaluru-560004
[Affiliated to Bengaluru City University]

LABORATORY REPORT
Of
"MOBILE APPLICATION DEVELOPMENT LABORATORY"

Submitted By:

NAME
UUCMS NO.

DEPARTMENT OF COMPUTER SCIENCE
THE NATIONAL COLLEGE
AUTONOMOUS
BASAVANAGUDI, BENGALURU - 560004

YEAR - 2026

The National College
Autonomous
Basavanagudi, Bengaluru-560004

Certificate

This is to certify that Mr. / Mrs.

NAME

UUCMS NO.

a student of class **Sixth Semester BCA,** has successfully completed the
"Mobile Application Development Laboratory"
practical work under the guidance of the **Department of Computer Science,** in partial fulfillment of the requirements for the award of the **Bachelor of Computer Applications** degree by
The National College, Basavanagudi, Bengaluru – 560 004
During the academic year 2026

Examiner's Signature

Teacher In Charge

Principal/HOD

Date :

Institution Seal

The candidate is expected to retain his / her report till he / she passes in the subject

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Mobile Application Development
Lab Programs

Program 1: Hello World Application

Create a simple Android application to display the message "Hello World" on the screen.

activity_main.xml

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        android:textSize="30sp"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

Java

```java
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Output:

Program 2: Screen Orientation Handling

Develop an application that displays different messages based on screen orientation changes (portrait and landscape).

layout/activity_main.xml (Portrait)

XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center">

  <TextView
    android:id="@+id/textView"
    android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="Portrait Mode"
        android:textSize="24sp" />

</LinearLayout>
```

layout-land/activity_main.xml (Landscape)

XML

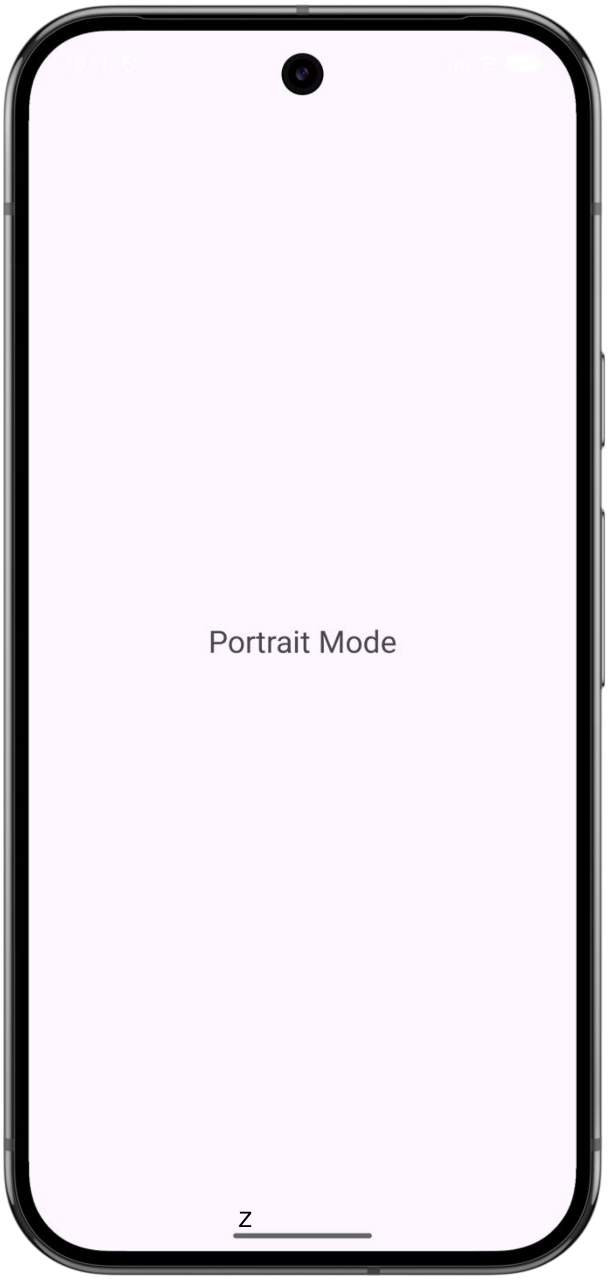
```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Landscape Mode"
        android:textSize="24sp" />

</LinearLayout>
```

MainActivity.java

Java

```java
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
```

Output:


Portrait Mode
Z


Landscape Mode

Program 3: Login Interface using UI Controls

Design a login window using basic UI controls such as TextView, EditText, and Button.

activity_main.xml *(Or activity_login.xml)*

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center"
  android:orientation="vertical"
  android:padding="16dp">

  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Login"
    android:textSize="24sp"
    android:textStyle="bold" />

  <EditText
    android:id="@+id/editTextUsername"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:hint="Username" />

  <EditText
    android:id="@+id/editTextPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="8dp"
    android:hint="Password"
    android:inputType="textPassword" />

  <Button
    android:id="@+id/buttonLogin"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:text="Login" />

</LinearLayout>
```

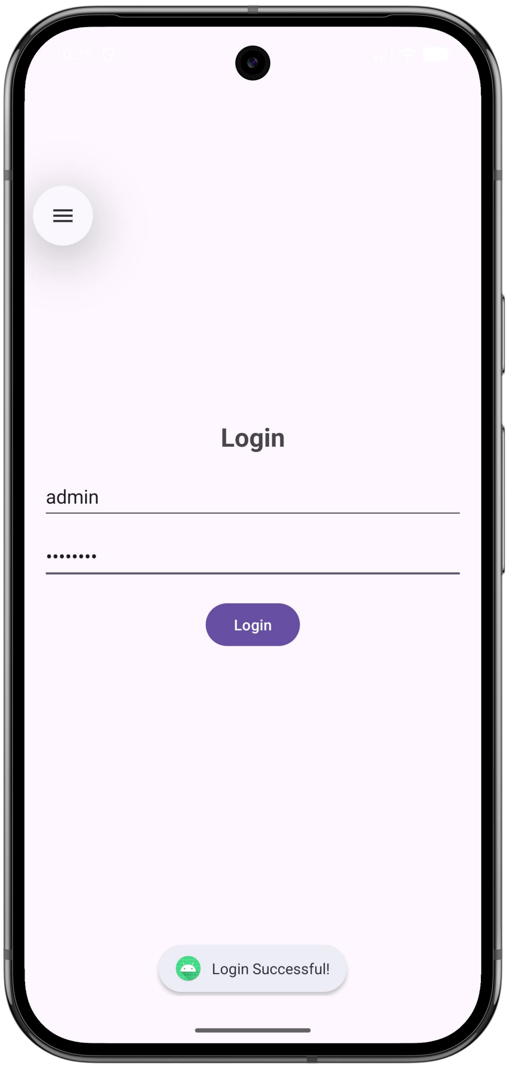
MainActivity.java *(Or LoginActivity.java)*

Java

```java
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
```

```java
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main); // Ensure this matches your XML file name

        EditText editTextUsername = findViewById(R.id.editTextUsername);
        EditText editTextPassword = findViewById(R.id.editTextPassword);
        Button buttonLogin = findViewById(R.id.buttonLogin);

        buttonLogin.setOnClickListener(v -> {
            String username = editTextUsername.getText().toString();
            String password = editTextPassword.getText().toString();

            if (username.equals("admin") && password.equals("password")) {
                Toast.makeText(MainActivity.this, "Login Successful!", Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(MainActivity.this, "Invalid Credentials!", Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

Output:



Program 4: Explicit and Implicit Intents

Create an application to demonstrate explicit and implicit intents for activity navigation and invoking built-in applications.

activity_main.xml

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center"
  android:orientation="vertical">

  <Button
    android:id="@+id/btnExplicit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Open Second Activity" />

  <Button
    android:id="@+id/btnDial"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:text="Open Dialer" />

  <Button
    android:id="@+id/btnBrowser"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:text="Open Browser" />

  <Button
    android:id="@+id/btnView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:text="Open Map" />

</LinearLayout>
```

MainActivity.java

Java

```java
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
```

```java
public class MainActivity extends AppCompatActivity {
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      Button btnExplicit = findViewById(R.id.btnExplicit);
      Button btnDial = findViewById(R.id.btnDial);
      Button btnBrowser = findViewById(R.id.btnBrowser);
      Button btnView = findViewById(R.id.btnView);

      // Explicit Intent
      btnExplicit.setOnClickListener(v -> {
         Intent intent = new Intent(MainActivity.this, SecondActivity.class);
         startActivity(intent);
      });

      // Implicit Intent - Dialer
      btnDial.setOnClickListener(v -> {
         Intent intent = new Intent(Intent.ACTION_DIAL);
         intent.setData(Uri.parse("tel:9876543210"));
         startActivity(intent);
      });

      // Implicit Intent - Browser
      btnBrowser.setOnClickListener(v -> {
         Intent intent = new Intent(Intent.ACTION_VIEW);
         intent.setData(Uri.parse("https://www.google.com"));
         startActivity(intent);
      });

      // Implicit Intent - Map
      btnView.setOnClickListener(v -> {
         Intent intent = new Intent(Intent.ACTION_VIEW);
         intent.setData(Uri.parse("geo:28.7041,77.1025"));
         startActivity(intent);
      });
   }
}
```

activity_second.xml

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:gravity="center"
   android:orientation="vertical">

   <TextView
```

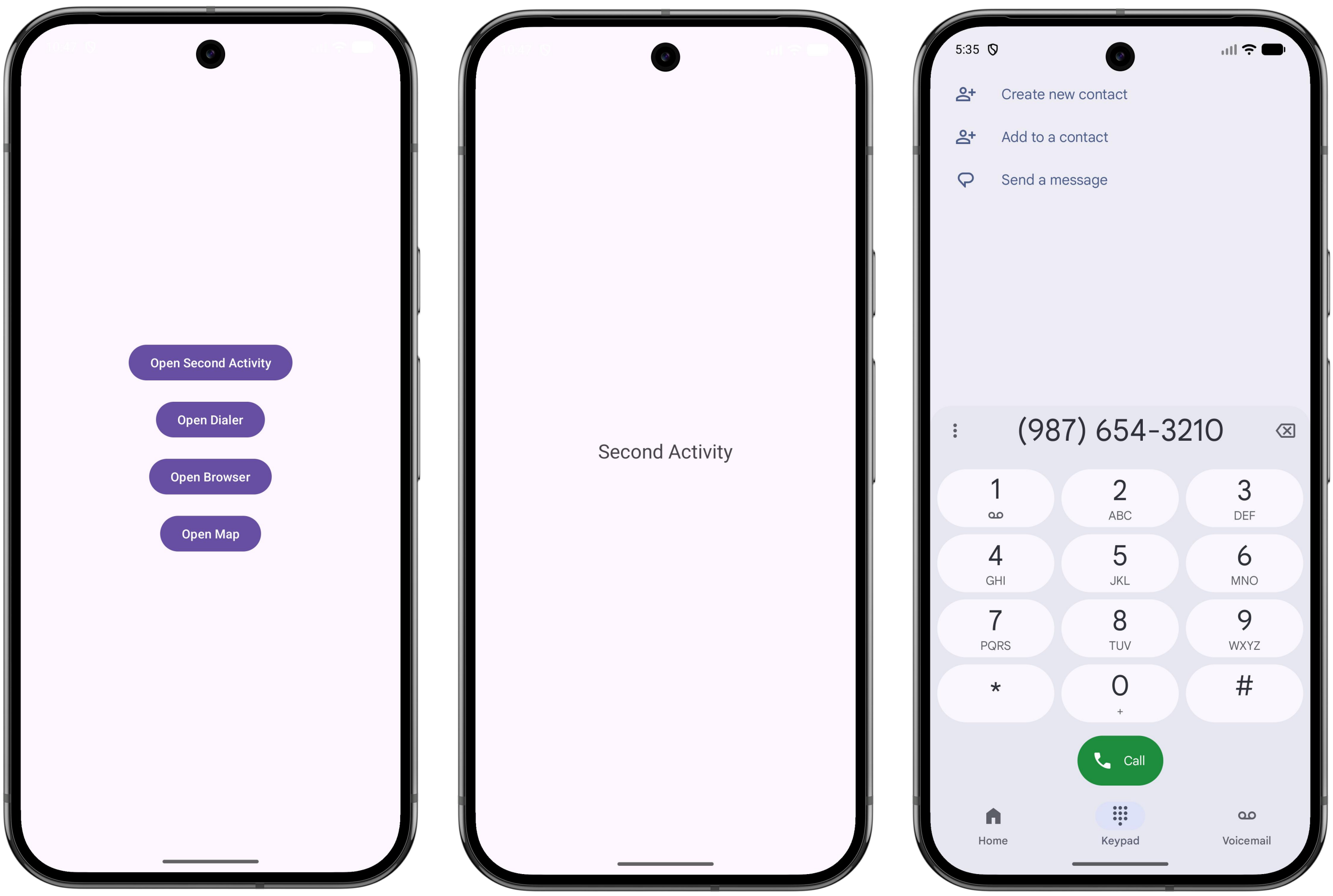
```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Second Activity"
    android:textSize="22sp"/>

</LinearLayout>
```

SecondActivity.java

Java

```java
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_second);
  }
}
```

Output:



Program 5: Custom Opening (Splash) Screen

Develop an application with a custom-designed opening screen displayed for a short duration before the main activity.

(Note: To make the Splash Screen open first, remember to move the <intent-filter> launcher block into .SplashActivity inside your AndroidManifest.xml!)

activity_splash.xml

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center">

  <ImageView
    android:id="@+id/logo"
    android:layout_width="150dp"
    android:layout_height="150dp"
    android:src="@mipmap/ic_launcher"
    android:layout_centerInParent="true"
    android:contentDescription="Logo" />

  <TextView
    android:id="@+id/appName"
    android:layout_below="@id/logo"
    android:layout_marginTop="20dp"
    android:layout_centerHorizontal="true"
    android:text="My Splash App"
    android:textColor="@color/black"
    android:textSize="24sp"
    android:textStyle="bold"
    android:layout_height="wrap_content"
    android:layout_width="wrap_content"/>

</RelativeLayout>
```

SplashActivity.java

Java

```java
import android.content.Intent;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.view.WindowManager;
import androidx.appcompat.app.AppCompatActivity;

public class SplashActivity extends AppCompatActivity {
  private static final int SPLASH_SCREEN_TIME_OUT = 2000;

  @Override
```

```java
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        // Make activity full screen
        getWindow().setFlags(WindowManager.LayoutParams.FLAG_FULLSCREEN,
                WindowManager.LayoutParams.FLAG_FULLSCREEN);

        setContentView(R.layout.activity_splash);

        new Handler(Looper.getMainLooper()).postDelayed(() -> {
            Intent i = new Intent(SplashActivity.this, MainActivity.class);
            startActivity(i);
            finish();
        }, SPLASH_SCREEN_TIME_OUT);
    }
}
```

activity_main.xml *(The screen that loads after the splash)*

XML

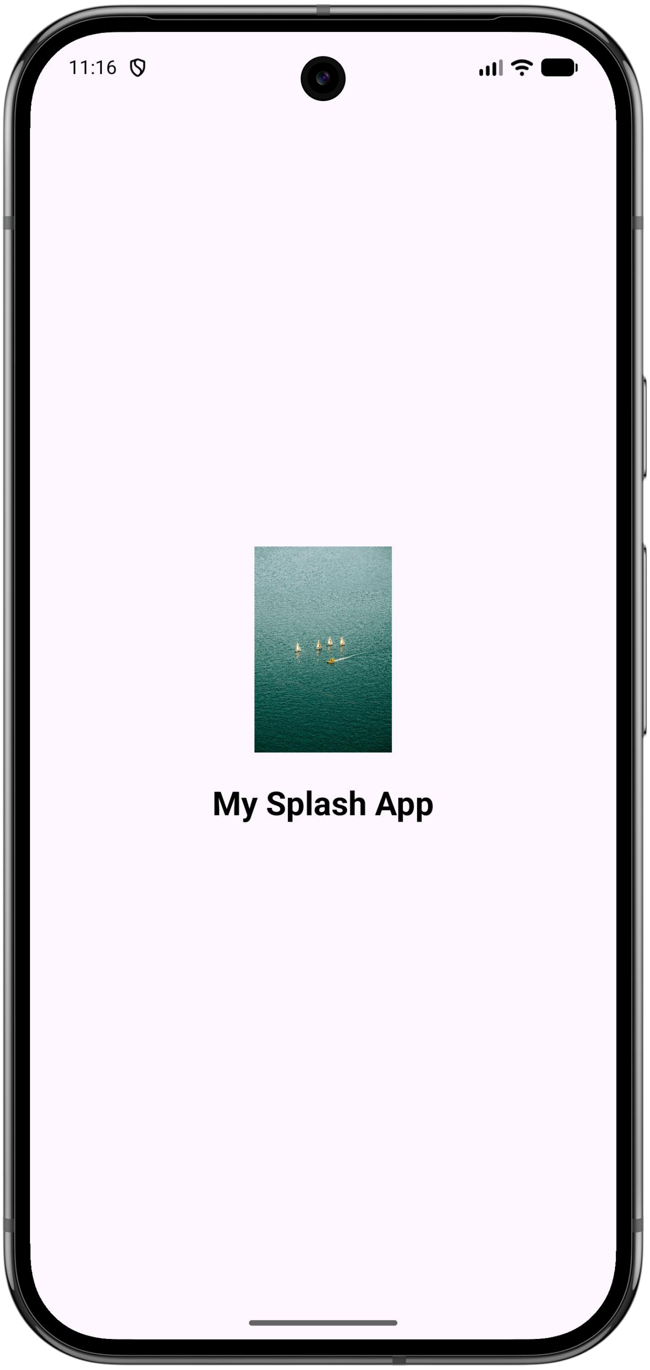
```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to the Main Screen!"
        android:textColor="#616161"
        android:textSize="24sp"
        android:textStyle="bold" />

</LinearLayout>
```

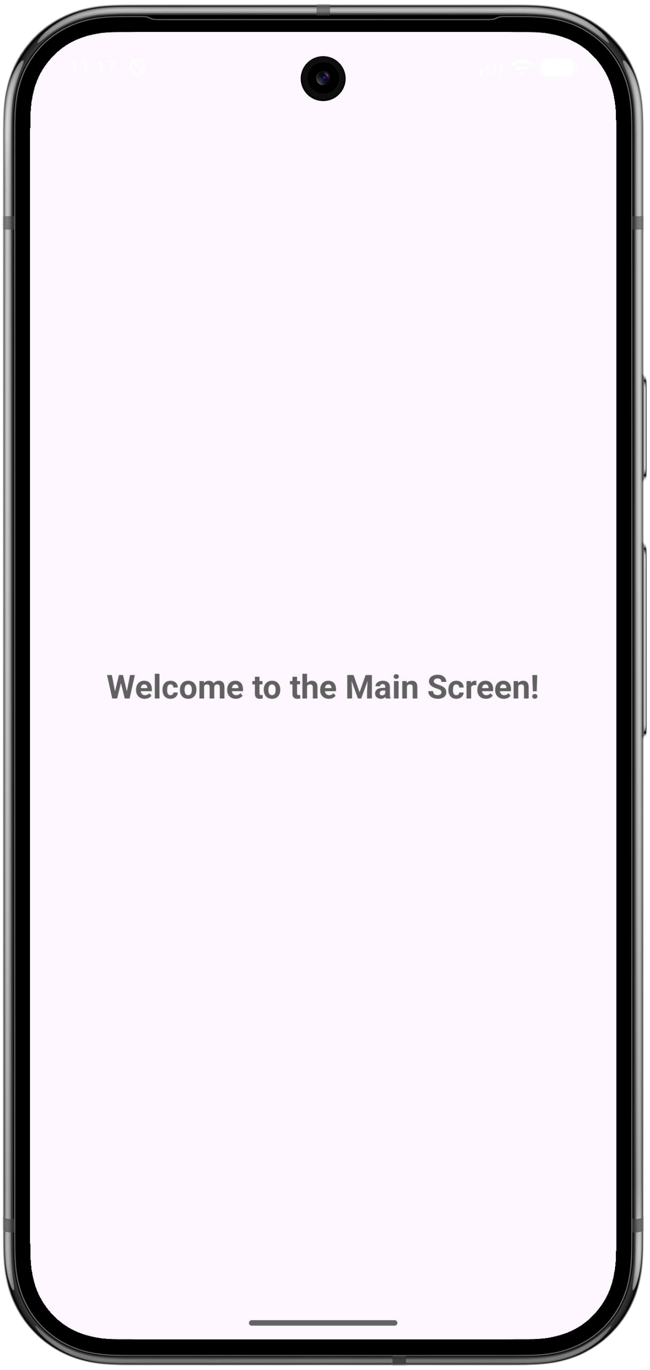
MainActivity.java

Java

```java
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Output:





Program 6: UI Design using Layouts and Views

Design a user interface using layouts and basic views such as LinearLayout / ConstraintLayout, TextView, EditText, Button, and ImageView.

activity_main.xml

XML

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  android:layout_width="match_parent"
  android:layout_height="match_parent">
```

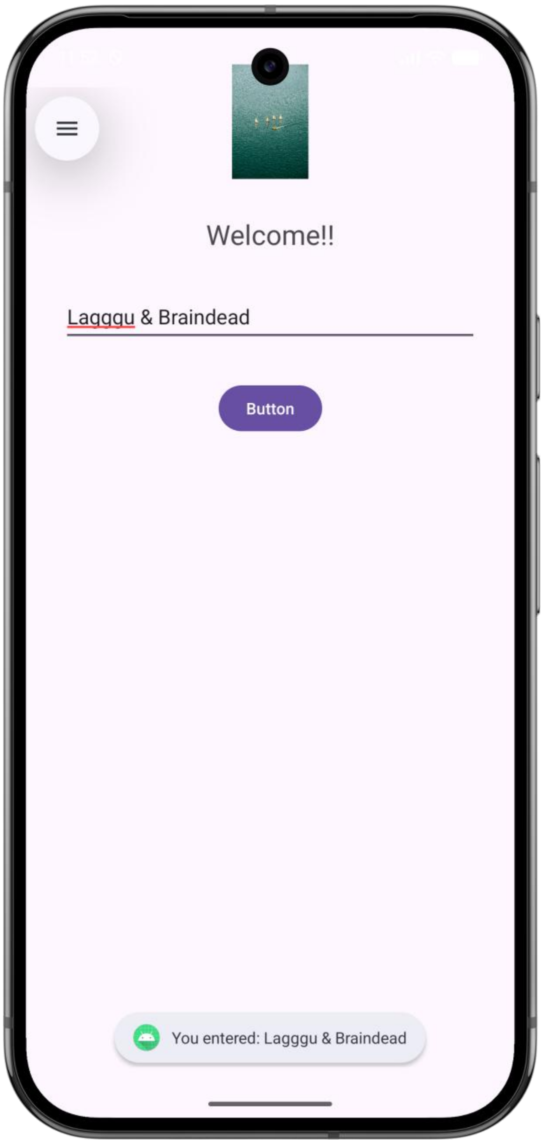
```xml
    <ImageView
        android:id="@+id/imageView"
        android:layout_width="100dp"
        android:layout_height="100dp"
        android:layout_marginTop="32dp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        android:src="@mipmap/ic_launcher" />

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="32dp"
        android:text="Welcome!!"
        android:textSize="24sp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/imageView" />

    <EditText
        android:id="@+id/editText"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginStart="32dp"
        android:layout_marginTop="32dp"
        android:layout_marginEnd="32dp"
        android:hint="Enter some text"
        android:inputType="text"
        android:minHeight="48dp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView" />

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="32dp"
        android:text="Button"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/editText" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

Java

```java
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        EditText editText = findViewById(R.id.editText);
        Button button = findViewById(R.id.button);

        button.setOnClickListener(v -> {
            String text = editText.getText().toString();
            Toast.makeText(MainActivity.this, "You entered: " + text, Toast.LENGTH_SHORT).show();
        });
    }
}
```

Output:



Program 7: Menus and Toast Notifications

Create an application that demonstrates the use of options menu and toast notifications.

activity_main.xml

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center"
  android:orientation="vertical">

  <TextView
    android:text="Welcome to the Menu App"
    android:textSize="20sp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

</LinearLayout>
```

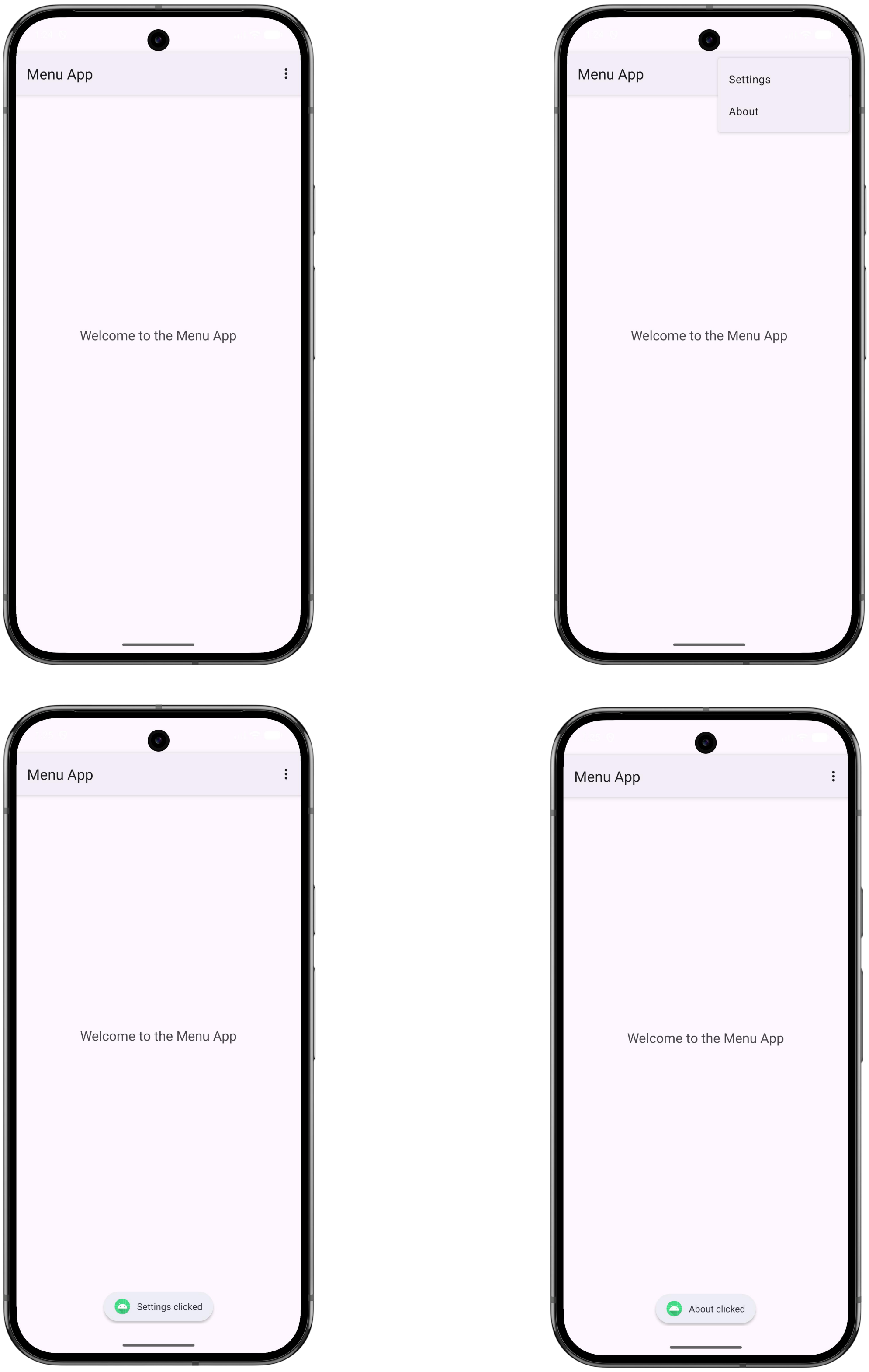
MainActivity.java

Java

```java
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
  }

  @Override
  public boolean onCreateOptionsMenu(Menu menu) {
    createMenu(menu);
    return true;
  }

  private void createMenu(Menu menu) {
    menu.add(0, 1, Menu.NONE, "Settings");
    menu.add(0, 2, Menu.NONE, "About");
  }

  @Override
  public boolean onOptionsItemSelected(MenuItem item) {
    switch (item.getItemId()) {
      case 1:
```

```
                Toast.makeText(this, "Settings clicked", Toast.LENGTH_SHORT).show();
                return true;
            case 2:
                Toast.makeText(this, "About clicked", Toast.LENGTH_SHORT).show();
                return true;
            default:
                return super.onOptionsItemSelected(item);
        }
    }
}
```

Output:



Program 8: Local Data Storage using Shared Preferences

Develop an application to store and retrieve data locally using Shared Preferences.

activity_main.xml

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:padding="16dp">

  <TextView
    android:id="@+id/textViewTitle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="SharedPreferences Demo"
    android:textSize="24sp"
    android:textStyle="bold"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

  <EditText
    android:id="@+id/editTextName"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginTop="32dp"
    android:hint="Enter your name"
    android:inputType="textPersonName"
    android:minHeight="48dp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@id/textViewTitle" />

  <Button
    android:id="@+id/buttonSave"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:text="Save Name"
    app:layout_constraintEnd_toStartOf="@+id/buttonClear"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@id/editTextName" />

  <Button
    android:id="@+id/buttonClear"
    android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="Clear Name"
        app:layout_constraintBottom_toBottomOf="@+id/buttonSave"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toEndOf="@+id/buttonSave"
        app:layout_constraintTop_toTopOf="@+id/buttonSave" />

    <TextView
        android:id="@+id/textViewSavedValue"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="32dp"
        android:text="No name saved yet"
        android:textSize="18sp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/buttonSave" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

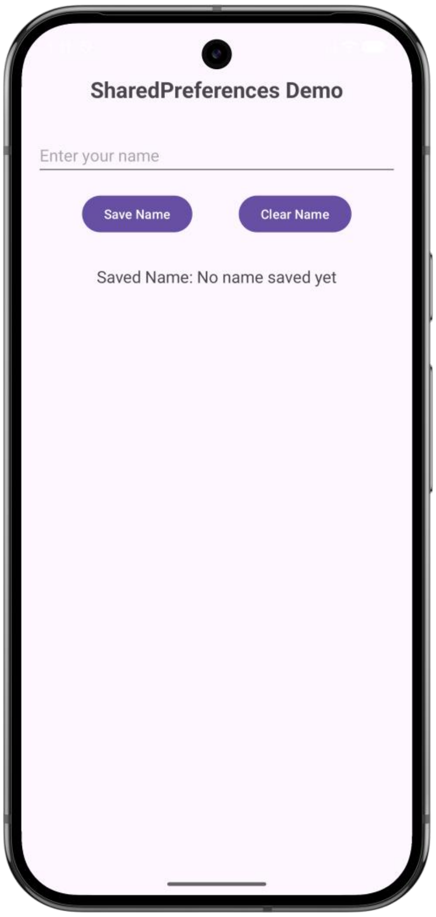
MainActivity.java

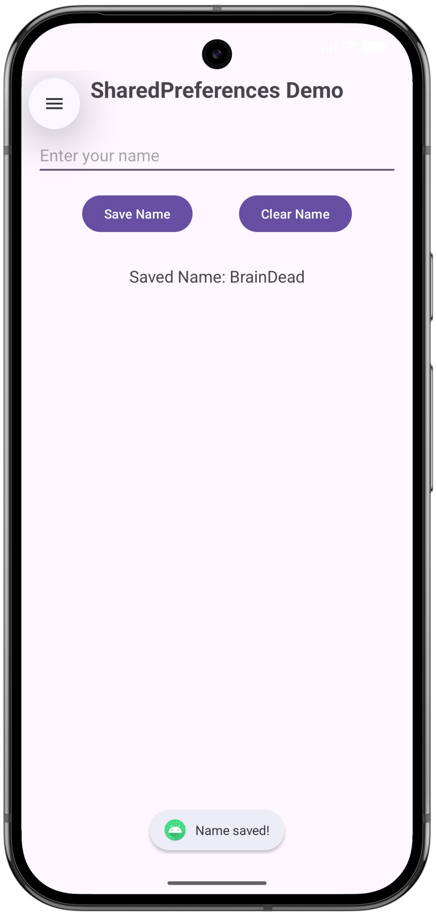
Java

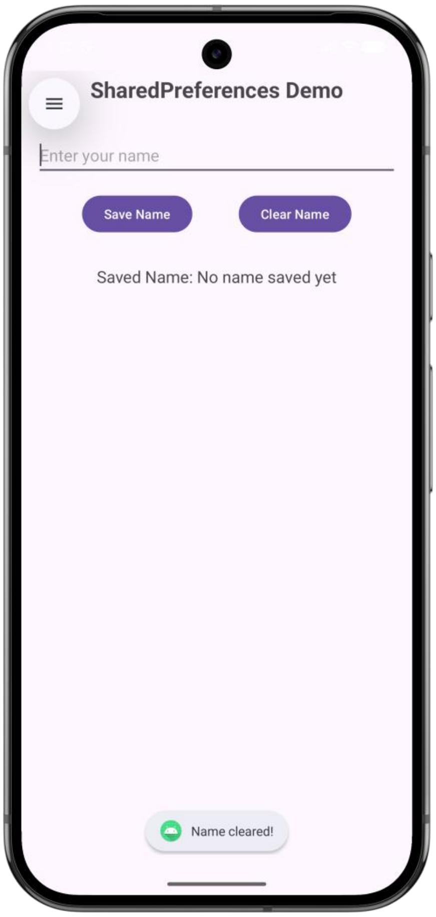
```java
import android.content.Context;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    private static final String PREFS_NAME = "MyPrefsFile";
    private static final String KEY_NAME = "user_name";

    private EditText editTextName;
    private TextView textViewSavedValue;
    private SharedPreferences sharedPreferences;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editTextName = findViewById(R.id.editTextName);
        textViewSavedValue = findViewById(R.id.textViewSavedValue);
        Button buttonSave = findViewById(R.id.buttonSave);
        Button buttonClear = findViewById(R.id.buttonClear);

        sharedPreferences = getSharedPreferences(PREFS_NAME, Context.MODE_PRIVATE);
```

```
        displaySavedName();

        buttonSave.setOnClickListener(v -> {
            String name = editTextName.getText().toString().trim();
            if (!name.isEmpty()) {
                saveName(name);
                displaySavedName();
                editTextName.setText("");
                Toast.makeText(MainActivity.this, "Name saved!", Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(MainActivity.this, "Please enter a name", Toast.LENGTH_SHORT).show();
            }
        });

        buttonClear.setOnClickListener(v -> {
            clearName();
            displaySavedName();
            Toast.makeText(MainActivity.this, "Name cleared!", Toast.LENGTH_SHORT).show();
        });
    }

    private void saveName(String name) {
        SharedPreferences.Editor editor = sharedPreferences.edit();
        editor.putString(KEY_NAME, name);
        editor.apply();
    }

    private void displaySavedName() {
        String savedName = sharedPreferences.getString(KEY_NAME, "No name saved yet");
        textViewSavedValue.setText("Saved Name: " + savedName);
    }

    private void clearName() {
        SharedPreferences.Editor editor = sharedPreferences.edit();
        editor.remove(KEY_NAME);
        editor.apply();
    }
}
```

Output:







Program 9: Database Operations using Room

Create an application to insert and display data using Room database.

(Remember to add Room dependencies: implementation("androidx.room:room-runtime:2.6.1") and annotationProcessor("androidx.room:room-compiler:2.6.1") in your app-level build.gradle.kts file and sync your project!)

activity_main.xml

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter a name to save" />

    <Button
```

```
        android:id="@+id/btnSave"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="Save to Room Database" />

    <Button
        android:id="@+id/btnClear"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp"
        android:backgroundTint="#D32F2F"
        android:textColor="#FFFFFF"
        android:text="Clear Database" />

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp"
        android:text="Database Entries:"
        android:textStyle="bold"
        android:textSize="18sp" />

    <TextView
        android:id="@+id/tvData"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp"
        android:textSize="16sp" />

</LinearLayout>
```

User.java *(Entity)*

Java

```java
import androidx.room.Entity;
import androidx.room.PrimaryKey;

@Entity(tableName = "users_table")
public class User {

    @PrimaryKey(autoGenerate = true)
    public int id;

    public String name;

    public User(String name) {
        this.name = name;
    }
}
```

UserDao.java *(DAO - Interface)*

```java
import androidx.room.Dao;
import androidx.room.Insert;
import androidx.room.Query;
import java.util.List;

@Dao
public interface UserDao {

    @Insert
    void insert(User user);

    @Query("SELECT * FROM users_table")
    List<User> getAllUsers();

    @Query("DELETE FROM users_table")
    void deleteAllUsers();
}
```

AppDatabase.java *(Database)*

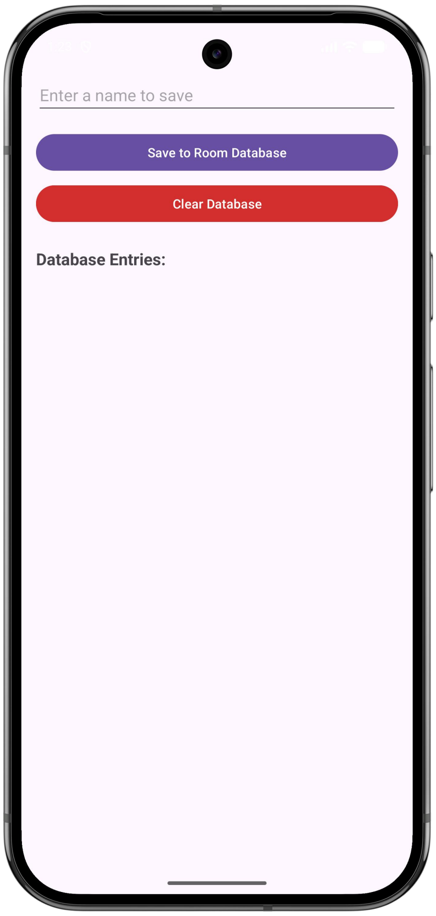
```java
import android.content.Context;
import androidx.room.Database;
import androidx.room.Room;
import androidx.room.RoomDatabase;

@Database(entities = {User.class}, version = 1)
public abstract class AppDatabase extends RoomDatabase {

    public abstract UserDao userDao();

    private static volatile AppDatabase INSTANCE;

    public static AppDatabase getInstance(Context context) {
        if (INSTANCE == null) {
            synchronized (AppDatabase.class) {
                if (INSTANCE == null) {
                    INSTANCE = Room.databaseBuilder(context.getApplicationContext(),
                            AppDatabase.class, "user_database")
                            .allowMainThreadQueries()
                            .build();
                }
            }
        }
        return INSTANCE;
    }
}
```

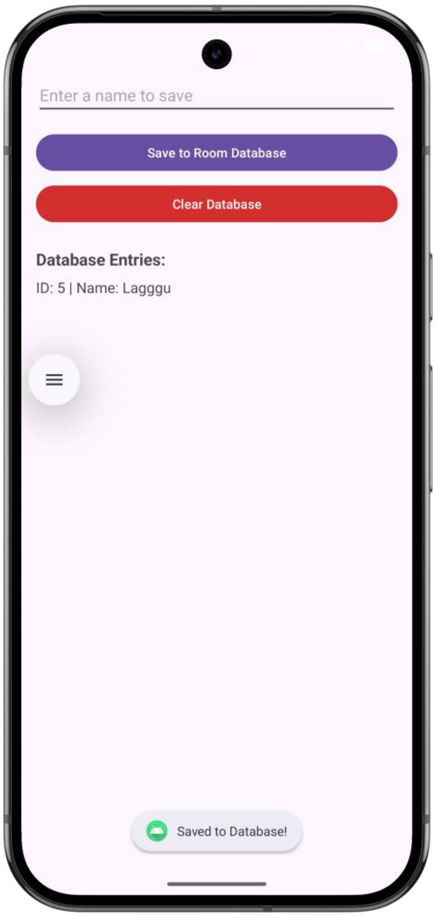
MainActivity.java

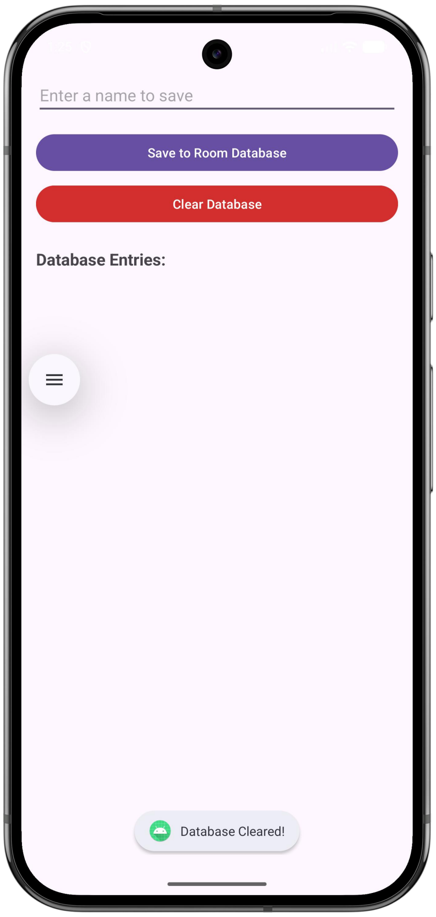
```java
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import java.util.List;

public class MainActivity extends AppCompatActivity {

    private AppDatabase db;
    private TextView tvData;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        EditText etName = findViewById(R.id.etName);
        Button btnSave = findViewById(R.id.btnSave);
        Button btnClear = findViewById(R.id.btnClear);
        tvData = findViewById(R.id.tvData);

        db = AppDatabase.getInstance(this);
        displayDatabaseInfo();

        btnSave.setOnClickListener(v -> {
            String name = etName.getText().toString().trim();

            if (!name.isEmpty()) {
                db.userDao().insert(new User(name));
                etName.setText("");
                displayDatabaseInfo();
                Toast.makeText(this, "Saved to Database!", Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "Please enter a name first", Toast.LENGTH_SHORT).show();
            }
        });

        btnClear.setOnClickListener(v -> {
            db.userDao().deleteAllUsers();
            displayDatabaseInfo();
            Toast.makeText(this, "Database Cleared!", Toast.LENGTH_SHORT).show();
        });
    }

    private void displayDatabaseInfo() {
        List<User> users = db.userDao().getAllUsers();
        StringBuilder sb = new StringBuilder();
```

```
        for (User u : users) {
            sb.append("ID: ").append(u.id).append(" | Name: ").append(u.name).append("\n");
        }

        tvData.setText(sb.toString());
    }
}
```

Output:







Program 10: Material Design

Develop an application to demonstrate material design.

activity_main.xml

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:background="#F5F5F5">

  <com.google.android.material.appbar.AppBarLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content">

    <com.google.android.material.appbar.MaterialToolbar
      android:id="@+id/topAppBar"
      android:layout_width="match_parent"
      android:layout_height="?attr/actionBarSize"
      app:title="Material Tasks"
      app:titleTextColor="@color/white"
      style="@style/Widget.MaterialComponents.Toolbar.Primary" />

  </com.google.android.material.appbar.AppBarLayout>

  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    app:layout_behavior="@string/appbar_scrolling_view_behavior">

    <com.google.android.material.card.MaterialCardView
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      app:cardElevation="4dp"
      app:cardCornerRadius="12dp"
      app:strokeWidth="1dp"
      app:strokeColor="#DDDDDD">

      <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <TextView
          android:layout_width="wrap_content"
```

```
            android:layout_height="wrap_content"
            android:text="Finish Design Project"
            android:textAppearance="?attr/textAppearanceHeadline6" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Due: Today, 5:00 PM"
            android:textAppearance="?attr/textAppearanceBody2"
            android:textColor="#757575" />

        <com.google.android.material.button.MaterialButton
            style="@style/Widget.MaterialComponents.Button.TextButton"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="8dp"
            android:text="MARK AS DONE" />

      </LinearLayout>
    </com.google.android.material.card.MaterialCardView>
  </LinearLayout>

  <com.google.android.material.floatingactionbutton.FloatingActionButton
    android:id="@+id/fab"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="bottom|end"
    android:layout_margin="24dp"
    android:contentDescription="Add Task"
    app:srcCompat="@android:drawable/ic_input_add" />

</androidx.coordinatorlayout.widget.CoordinatorLayout>
```

MainActivity.java

Java

```java
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import com.google.android.material.floatingactionbutton.FloatingActionButton;
import com.google.android.material.snackbar.Snackbar;

public class MainActivity extends AppCompatActivity {
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);
```

```
        // This part ensures the UI doesn't hide under the Status Bar
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        // Add functionality to your Floating Action Button
        FloatingActionButton fab = findViewById(R.id.fab);
        fab.setOnClickListener(view -> {
            Snackbar.make(view, "Task added successfully!", Snackbar.LENGTH_LONG)
                    .setAction("Undo", null)
                    .show();
        });
    }
}
```

Output:

Program 11: Fragment-based Navigation

Create a multi-screen application using Fragments, demonstrating fragment transactions.

fragment_first.xml

XML

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:background="#FFCDD2">

  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="center"
    android:text="First Fragment"
    android:textSize="24sp"
    android:textStyle="bold" />

</FrameLayout>
```

FirstFragment.java

Java

```
import android.os.Bundle;
import androidx.fragment.app.Fragment;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;

public class FirstFragment extends Fragment {
  public FirstFragment() {
    // Required empty public constructor
  }

  @Override
  public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState) {
    return inflater.inflate(R.layout.fragment_first, container, false);
  }
}
```

fragment_second.xml

XML

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:background="#C8E6C9">

  <TextView
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:text="Second Fragment"
        android:textSize="24sp"
        android:textStyle="bold" />

</FrameLayout>
```

SecondFragment.java

Java

```java
import android.os.Bundle;
import androidx.fragment.app.Fragment;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;

public class SecondFragment extends Fragment {
    public SecondFragment() {
        // Required empty public constructor
    }

    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState) {
        return inflater.inflate(R.layout.fragment_second, container, false);
    }
}
```

activity_main.xml

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:padding="8dp">

        <Button
            android:id="@+id/btnFragment1"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:layout_marginTop="25dp"
            android:text="Fragment 1" />
```

```
        <Button
            android:id="@+id/btnFragment2"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:layout_marginTop="25dp"
            android:text="Fragment 2"
            android:layout_marginStart="8dp"/>

    </LinearLayout>

    <FrameLayout
        android:id="@+id/fragmentContainer"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

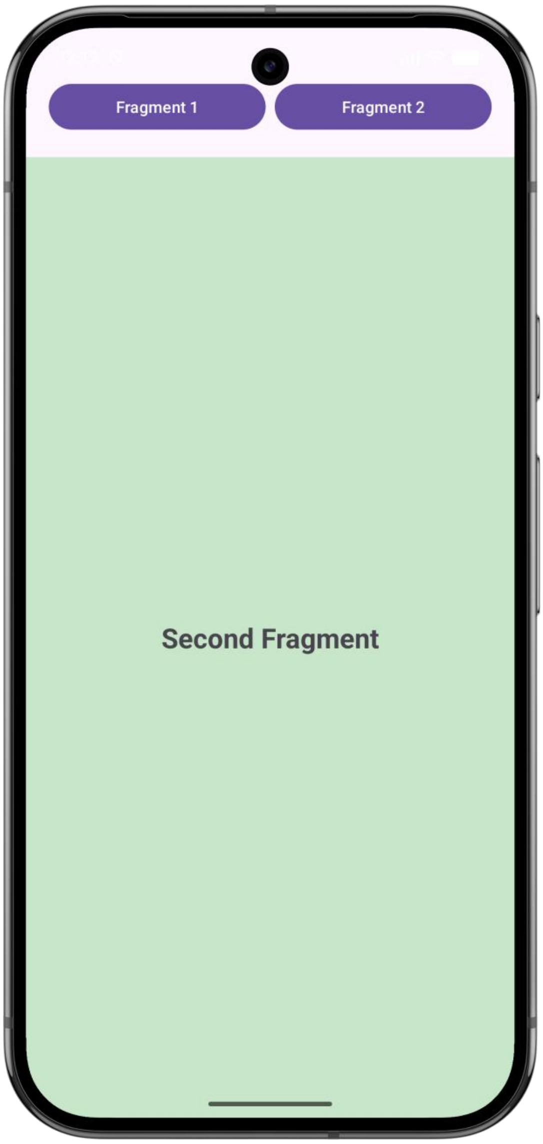
MainActivity.java

Java

```java
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentTransaction;

public class MainActivity extends AppCompatActivity {
    Button btnFragment1, btnFragment2;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnFragment1 = findViewById(R.id.btnFragment1);
        btnFragment2 = findViewById(R.id.btnFragment2);

        // Load default fragment
        loadFragment(new FirstFragment());

        btnFragment1.setOnClickListener(v -> loadFragment(new FirstFragment()));
        btnFragment2.setOnClickListener(v -> loadFragment(new SecondFragment()));
    }

    private void loadFragment(Fragment fragment) {
        FragmentManager fm = getSupportFragmentManager();
        FragmentTransaction ft = fm.beginTransaction();
        ft.replace(R.id.fragmentContainer, fragment);
        ft.commit();
```

```
    }
}
```

Output:



Program 12: Action Bar

Create an application that demonstrates the implementation and use of ActionBar.

Step 1: Unhide the Action Bar in themes.xml

By default, new Android Studio projects hide the Action Bar. To make it visible for this program:

Go to app > res > values > themes.xml (and also themes.xml (night) if applicable).
Find the <style> tag for your base theme.
Remove .NoActionBar from the end of the parent attribute.
Example: Change parent="Theme.Material3.DayNight.NoActionBar" to parent="Theme.Material3.DayNight".

activity_main.xml

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="24dp">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Action Bar Demonstration"
        android:textSize="24sp"
        android:textStyle="bold"
        android:textAlignment="center"
        app:layout_constraintBottom_toTopOf="@+id/textView2"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_chainStyle="packed" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:layout_marginStart="32dp"
        android:layout_marginEnd="32dp"
        android:text="Use the three-dot menu in the Action Bar above to navigate to the Second Activity."
        android:textSize="16sp"
        android:textAlignment="center"
        android:textColor="#666666"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

menu/main_menu.xml

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">
    <item
        android:id="@+id/action_search"
        android:icon="@android:drawable/ic_menu_search"
        android:title="Search"
        app:showAsAction="ifRoom" />
    <item
```

```
    android:id="@+id/action_next_page"
    android:title="Open Second Activity"
    app:showAsAction="never" />
  <item
    android:id="@+id/action_settings"
    android:title="Settings"
    app:showAsAction="never" />
  <item
    android:id="@+id/action_about"
    android:title="About"
    app:showAsAction="never" />
</menu>
```

MainActivity.java

Java

```java
import android.content.Intent;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    // Set the title of the Action Bar
    if (getSupportActionBar() != null) {
      getSupportActionBar().setTitle(R.string.app_name);
    }
  }

  @Override
  public boolean onCreateOptionsMenu(Menu menu) {
    // Inflate the menu; this adds items to the action bar if it is present.
    getMenuInflater().inflate(R.menu.main_menu, menu);
    return true;
  }

  @Override
  public boolean onOptionsItemSelected(@NonNull MenuItem item) {
    int id = item.getItemId();

    if (id == R.id.action_search) {
      Toast.makeText(this, "Search clicked", Toast.LENGTH_SHORT).show();
      return true;
    } else if (id == R.id.action_next_page) {
      // Navigate via the Action Bar Menu
```

```
            Intent intent = new Intent(MainActivity.this, SecondActivity.class);
            startActivity(intent);
            return true;
        } else if (id == R.id.action_settings) {
            Toast.makeText(this, "Settings clicked", Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.action_about) {
            Toast.makeText(this, "About clicked", Toast.LENGTH_SHORT).show();
            return true;
        }

        return super.onOptionsItemSelected(item);
    }
}
```

activity_second.xml

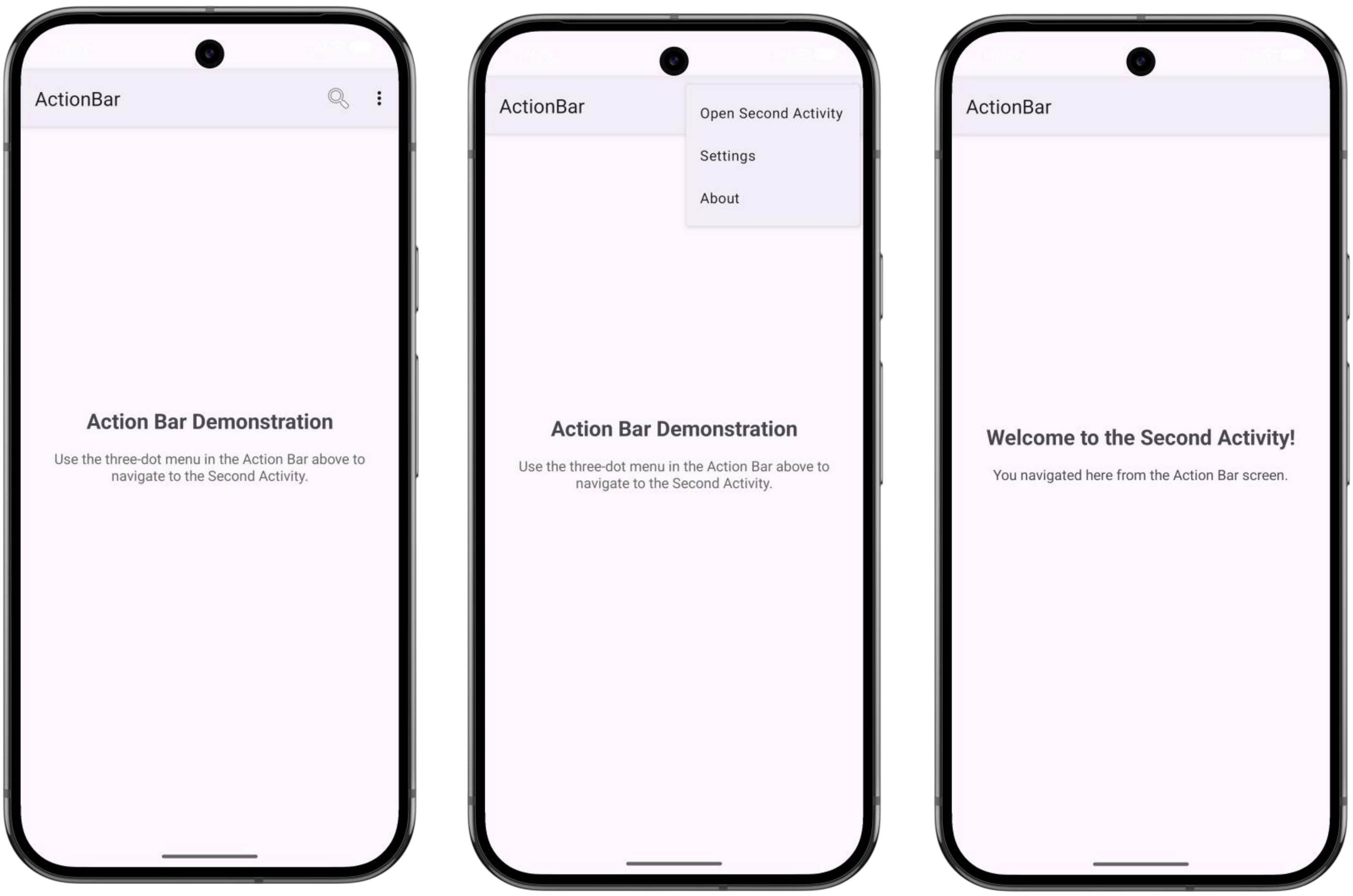
```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to the Second Activity!"
        android:textSize="24sp"
        android:textStyle="bold" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="You navigated here from the Action Bar screen."
        android:textSize="16sp" />

</LinearLayout>
```

SecondActivity.java

```java
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
```

```
        // Add a back arrow to the Action Bar on this second screen
        if (getSupportActionBar() != null) {
            getSupportActionBar().setTitle("Second Activity");
            getSupportActionBar().setDisplayHomeAsUpEnabled(true);
        }
    }

    // This makes the back arrow in the Action Bar actually work
    @Override
    public boolean onSupportNavigateUp() {
        finish();
        return true;
    }
}
```

Output:



Program 13: Android App Bundle (AAB) Generation

Demonstrate the steps to build and generate an Android App Bundle (AAB) for deployment using Android Studio.

Step 1: Open the Build Wizard

1. Launch Android Studio and open your project.
2. In the top menu bar, go to Build > Generate Signed Bundle / APK...
3. In the dialog that appears, select **Android App Bundle** and click **Next**.

Step 2: Set Up Signing Credentials

To upload to the Play Store, your bundle must be signed with a digital certificate.

If you already have a keystore:

1. Click Choose existing... and locate your .jks or .keystore file.
2. Enter your Key store password, Key alias, and Key password.

If you need a new keystore:

1. Click Create new...
2. Choose a key store path (somewhere safe outside your project folder).
3. Set passwords for both the Keystore and the Key.
4. Fill in at least one field in the Certificate section (e.g., your name).
5. Click **OK**.

Step 3: Configure Build Variants

1. After entering your credentials, click **Next**.
2. Select the Destination Folder where you want the AAB file to be saved.
3. Choose the Build Variant:
   - **release:** Use this for publishing to the Play Store.
   - **debug:** Use this only for internal testing.
4. Click **Finish**.

Step 4: Locate and Verify

1. Android Studio will begin the build process. You can monitor progress in the Build tab at the bottom.
2. Once complete, a pop-up notification will appear in the bottom-right corner.
3. Click Locate in that notification to open the folder containing your new .aab file.
   - *Default path:* project-name/module-name/release/app-release.aab